Task 3 — OCR Quality & Data Quality Analysis

OCR quality comparison before vs after preprocessing and data quality issues affecting quiz generation

1. Compare OCR Quality Before vs After Preprocessing

Document	Before Preprocessing	After Preprocessing	Difference	Result
C1	92.81%	89.75%	–3.06 pp	Decreased
C2	94.37%	94.49%	+0.12 pp	Increased
C3	86.23%	83.25%	–2.98 pp	Decreased
K1	87.83%	86.08%	–1.75 pp	Decreased
K2	83.12%	74.99%	–8.13 pp	Decreased
K3	85.68%	79.03%	–6.65 pp	Decreased

Overall Comparison

1. C2 was the only document where preprocessing improved OCR confidence, increasing from 94.37% to 94.49% (+0.12 percentage points).

2. C1, C3, K1, K2, and K3 showed a decrease in OCR confidence after preprocessing.

3. The largest decrease was observed for K2 (–8.13 percentage points), followed by K3 (–6.65 percentage points).

4. C3 decreased from 86.23% to 83.25%, a change of –2.98 percentage points.

5. Overall, basic preprocessing does not necessarily improve OCR for every document; its effectiveness depends on the document's content and image characteristics.

Conclusion: Preprocessing improved only C2 in this sample set, while the remaining five documents experienced lower OCR confidence.

2. Data Quality Issues That Can Affect Quiz Generation

The following data quality issues in OCR/scanned documents can directly affect automatic quiz generation:

1. OCR Character Errors

Incorrectly recognized letters, numbers, or symbols can change the meaning of the source content and lead to incorrect questions or answers.

2. Mathematical / Equation Errors

Subscripts, superscripts, operators, Greek symbols, and equations may be misread. This is especially important for C3 and K3, where mathematical content was difficult for OCR.

3. Table Structure Loss

OCR may extract table content but lose the correct rows, columns, and relationships. This can cause a quiz system to associate the wrong value with the wrong heading.

4. Missing or Incorrect Numbers

A single digit or decimal error can change a factual answer or numerical problem completely.

5. Special Symbol Errors

Symbols such as +, –, =, <, >, Greek letters, and scientific notation can be incorrectly recognized.

6. Diagram / Figure Information Loss

Normal OCR mainly extracts text. Important information contained in diagrams, labels, arrows, charts, or figures may be missed, resulting in incomplete quiz questions.

7. Reading Order / Layout Errors

Text from columns, boxes, headings, captions, or different sections may be combined in the wrong order. This can create questions using unrelated pieces of information.

8. Missing Text

If OCR fails to recognize a sentence, heading, footnote, or important keyword, the quiz generator may not know that information exists.

9. Duplicate Text

OCR can sometimes repeat text or extract overlapping content, which may lead to duplicate or repetitive quiz questions.

10. Incorrect Context

When OCR changes or rearranges content, a sentence may lose its original context. A quiz generator can then create a question whose answer is ambiguous or incorrect.

Key Impact on Quiz Generation: Poor OCR quality → inaccurate extracted text → incorrect or incomplete source information → incorrect, ambiguous, or incomplete quiz questions and answers.

Recommendation: OCR output should be checked for accuracy, especially for equations, numbers, tables, symbols, diagrams, and document layout, before it is used as input for automated quiz generation.